Team Guide: the 2 November update: how it works, how to test it, how to demo it

For: our own team (developers, testers, whoever gives the demo). **Covers:** what changed for the client's 8 points, how each flow works behind the screen, how to set up and test it, and how to run the demo. **Related:** `06b_SHORT_DEMO` (the click-by-click demo script), `06_CLIENT_DEMO_WALKTHROUGH` (the full 75-minute version), `docs/reports/CLIENT_REQUIREMENTS_ANALYSIS_2026-11-02.md` (§9–10: what was built and the decisions taken).

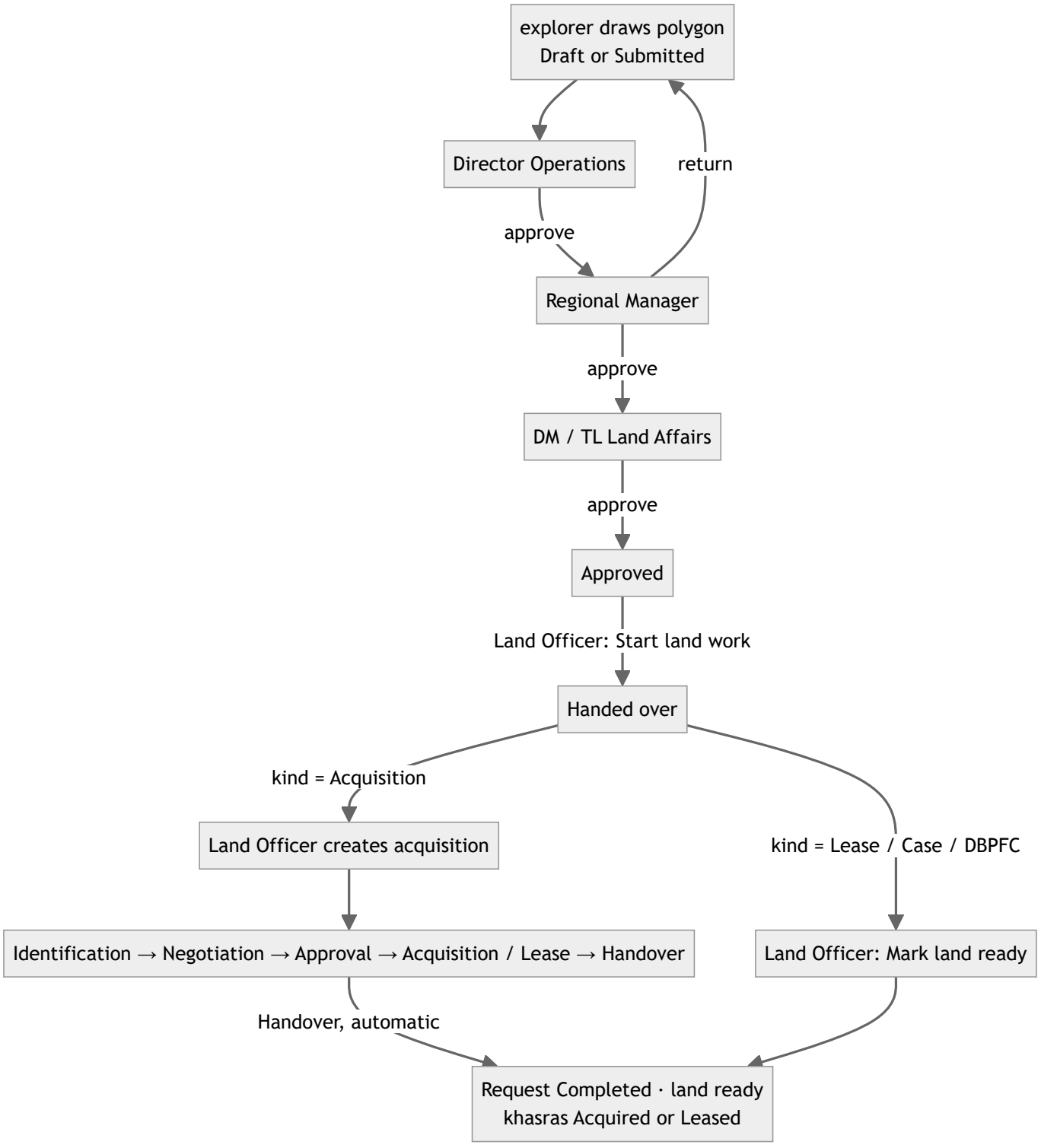
1. What is new, in one table

#	Client asked	What we built	Where on screen	Who uses it
1	Exploration request → approvals → purchase → progress → completion	Request follows its acquisition stage; acquisition **Handover** marks khasras Acquired/Leased and closes the request as **Completed (land ready)**; land requests in **Approvals & Alerts** with notices to the requester; lease form pre-filled from a request; **Mark land ready** for lease/case/DBPFC hand-overs	Land Requests, Approvals & Alerts, Acquisition & Title	explorer, dirops, regmgr, landaffairs, lofield
1	Temporary access	Role **"Valid until"** date: set, change, remove; access stops automatically after that day (Pakistan time)	Users & Roles → Access	admin
2	One account with all tabs	Demo account `director`: 33 tabs, read-only, CNIC masked	whole menu	director
3	No duplicate / wrong entries	Same CNIC blocks Save; similar name + father name warns; duplicate khasra refused with the existing code	Landowners, Survey / Khasra Register	lofield
4	Sold land → payments to new owner	Applying a mutation moves the seller's **unpaid** case share to the buyer/heirs (paid part stays; parcel-wise if he keeps other land in the case), adds them as lessors, and **holds** the seller's unreleased vouchers. Finance gets **Cancel voucher**	Mutation & Succession → Details; Payments & Cheques	lofield, sradmin, legal, mland, finance
5	Court case → payment hold	Already existed. New: the Legal Case form now has **Parcels concerned** and **Landowners concerned**; without them a stay held nothing	Legal & Litigation	legal, mland, finance
6	Acquisition demo	Now runs to the end (land ready)	—	—
7	Sindh hidden on mobile	Province chooser starts at the top on small screens	Select province screen	everyone
8	Province units, sizes in Settings	**Settings** → **Land units**: unit list per province and each local unit's size (per acre); khasra form shows the province's units first; 5 prototype purposes added	Settings, Survey / Khasra Register	admin, lofield

Database changes: migrations **0060**, **0061**, **0062**. Demo account seed: `db/seed/seed_director_account.sql`.

2. How each flow works

2.1 Land request → land ready



- **Who decides each tier:** the person holding that position (Director Operations, Regional Manager, DM / TL Land Affairs) in the request's province. The requester cannot approve; nobody approves two tiers. A return sends it back to the requester, and every tier decides again.
- **Progress for the requester:** each acquisition stage change is copied onto the request (`work_stage`). The explorer has no access to acquisitions but sees the strip in **Details**.

- **Khasra status at Handover:** Direct Purchase or LAA 1894 → **Acquired**. Private lease, government lease, ROW / easement → **Leased** (company has a right of use; title stays with the owner). Disputed or Released khasras are not touched.
- **Notices:** Approvals & Alerts → *Waiting on you* (approvers, Land Officer) and *Your land requests* (requester: returned, rejected, approved, handed over, land ready).

2.2 Temporary access

- Admin → **Users & Roles** → **Access** on the user → **Set end date** on the grant → date + reason.
- The date means the end of that day, Pakistan time. After it, the person has no permissions from that role; nothing has to be removed by hand.
- Technically a change is "revoke + new grant", so the history keeps both.

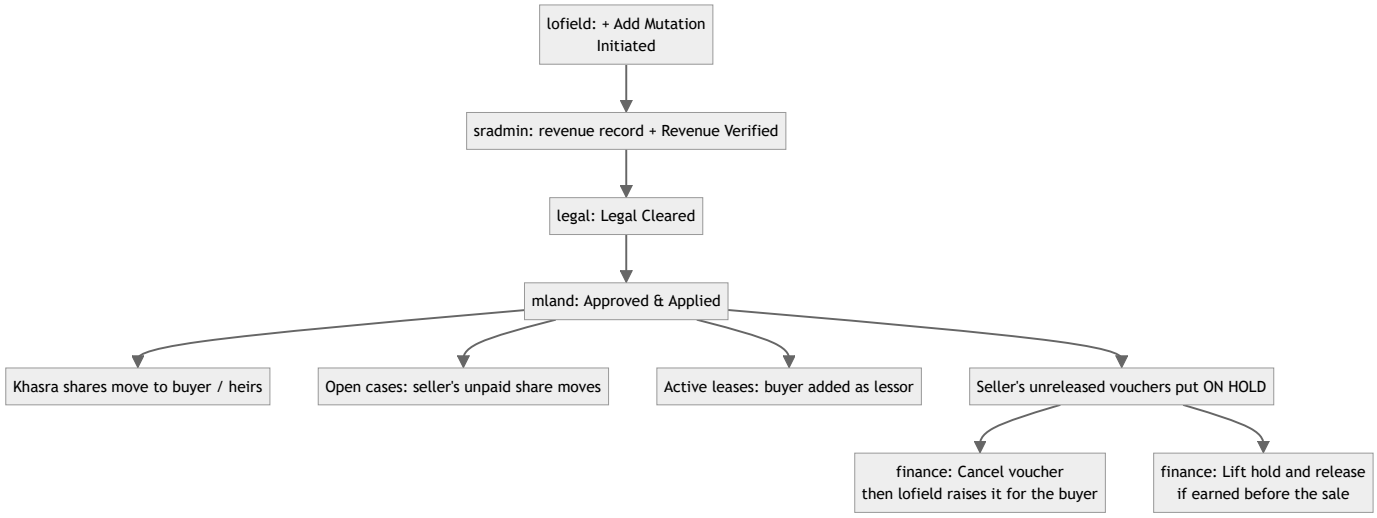
2.3 Province units

- **Settings → Land units → province tab.**
  - *Units used in …*: the order the khasra form shows them (**Change list…**).
  - *Unit sizes in …*: **Set size…** = how many of that unit make one acre, with a *confirmed* tick. **Use the national size** removes the override.
- Acre, hectare and square measures are fixed and cannot be resized.
- A new size applies to khasras saved from that day on. A khasra already recorded keeps its area.

2.4 Duplicates

Check	When	What happens
Same CNIC	Landowner form, on leaving the CNIC box	red box, **Save disabled**, Open button to the existing person
Similar name + father name in the province	on leaving the name boxes	amber warning only
Same khasra (survey no. + block + deh)	on Save	refused: "already exists in Jarwar as AREA-000034"

2.5 Mutation → payments



- **What the seller keeps on a case:** the part already paid to him. If he still owns another khasra in the same case, also the share for that khasra (worked out from the value of the case lines on each khasra).
- **Record:** Mutation → **Details** → **Payments re-pointed** lists every move.
- A new voucher for the seller on that case is refused ("not an owner on case …").

2.6 Court stay

- **Legal** records the case with the parcels and landowners concerned and ticks **Stay order**. The database places a hold on each one.
- Any voucher on those parcels or owners cannot be verified or released ("Payment is on hold: stay order …").
- **Manager Land** lifts it: Update → decided status, untick stay, type the court order. (Not the Legal Advisor: lifting releases money, so it needs payment-hold authority.)

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3. Set up before testing or a demo

Step	Command / action
1. Get the code	`git pull` (or use the working copy)
2. Migrate the database	`python db/tools/migrate.py --dsn "<lms_migrator DSN>" up`, then `status` must show 0 pending (62 applied)
3. Demo accounts	on a new database: `psql -U lms_migrator -d <db> -f db/seed/seed_exploration_accounts.sql` and `-f db/seed/seed_director_account.sql`; set passwords with `npx tsx scripts/iam/bootstrap.ts --username <name>` (or `npm run demo:setup` for a fresh demo DB)
4. Build and restart	`npm run build`, then restart (local: the 3040 server; live: `pm2 restart lms-app`)
5. Backup	`pg_dump -Fc -d <db> -f backups/before_demo.dump`
6. Browser	Chrome, 1440 × 900, one incognito window per account

Windows: set `PGCLIENTENCODING=UTF8` and `PYTHONUTF8=1` before migrate/psql.

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4. Test checklist (tick each; expected result in the last column)

Use the data and steps in `06b_SHORT_DEMO` (Part 0 creates the data).

#	Test	Account	Expected
T1	Phone view (F12 → device iPhone 12 Pro)	viewer	"Select province" heading and Sindh card at the top
T2	Menu sizes	director / explorer / viewer	33 / 9 / 7 tabs
T3	Director cannot change anything	director	no Add / Edit / approve buttons; CNIC masked
T4	Set role end date	admin	grant shows "until …"; past date refused

#	Test	Account	Expected
T5	Set Jireb size in Sindh	admin	row shows the new per-acre figure, "confirmed"
T6	Khasra unit list	lofield	unit dropdown: "Units of SND" first, then "Other units"
T7	Ghunta conversion	lofield	320 Ghunta saved as 8.00 ac
T8	Duplicate CNIC `45102-1100012-3`	lofield	red box, Save disabled
T9	Similar name `Mukhtar Ali Jarwar` / `Allah Ditto Jarwar`	lofield	amber warning naming SND-000002
T10	Duplicate khasra 451/2 block 9	lofield	refused with "already exists … AREA-000034"
T11	Request appears in approver's queue	dirops	Approvals & Alerts → Waiting on you
T12	Return goes back to requester	regmgr → explorer	explorer sees "Returned" notice; Edit on map
T13	Requester cannot approve own request	explorer	no Review button
T14	Hand-over marks khasras	lofield	"2 parcel(s) marked Under Acquisition"
T15	Acquisition pre-filled from request	lofield	project, area, khasras filled in
T16	Requester sees progress	explorer	Details → strip moves with each acquisition stage
T17	Handover completes the request	sradmin → explorer	request Completed; "Land ready" notice; khasras Leased
T18	Lease hand-over closed by hand	lofield	**Mark land ready** button (not offered for acquisition)
T19	Stay holds payment	legal → finance	Verify refused with "stay order" message
T20	Only Manager Land lifts the stay	legal → mland	legal refused; mland succeeds; Verify then works
T21	Mutation moves the case share	mland	case owners Ghulam 60 / Abdul Ghaffar 40; Details shows the move
T22	Seller's voucher held	finance	VCH of Nazir shows On hold (ownership changed)
T23	Cancel voucher	finance	voucher Cancelled; hold lifted
T24	Voucher for buyer	lofield	accepted, WHT 10% (Filer)
T25	Voucher for seller refused	lofield	"not an owner on case …"
T26	CNIC search	lofield	`45104-2000013-9` finds SND-000049
T27	Audit	audit	MUT-0007 changes listed with who and when

Automated tests (developers):

Suite	Command	Last result (2 Nov)
Typecheck, lint	`npm run typecheck` · `npm run lint`	pass
DB	`bash db/tools/rebuild.sh lms_test seed` then `python db/tools/run_tests.py --db lms_test --no-rebuild`	26/26, 919 checks

Suite	Command	Last result (2 Nov)
API	server on 3041 + `npx vitest run tests/iam tests/regression`	1545/1548; rerun of failures 109/109
Demo rehearsal	the whole walkthrough through the API on a DB copy	63 checks, 0 failures

5. Giving the demo

1. **Before:** set up (§3), enter Part 0 data from `06b_SHORT_DEMO`, take the backup, open incognito windows for the accounts you will switch between.
2. **Order** (follows the client's messages): all tabs → mobile → exploration request and approvals → acquisition and land ready → court stay → land sold → CNIC and audit. Units and temporary access fit after the tabs if time allows.
3. **One sentence per part** to say to the client:

Part	Say
Tabs	"One overview account sees everything and changes nothing; every other role sees only its own work."
Request	"Exploration marks the land on the map; three positions approve in turn; nobody approves their own or two tiers."
Land ready	"The requester sees each stage of the acquisition and is told when the land is ready."
Court stay	"A stay order stops every payment on that land and owner, in the database, until the court's order is recorded."
Land sold	"Field starts it, revenue and legal verify, head office applies; unpaid money moves to the new owner, and anything already raised for the old owner waits for Finance."
Duplicates	"The same CNIC or khasra can never be entered twice."
Units	"Each province has its own units and sizes, set here; old records keep their area."

1. **After:** to repeat, restore the backup and restart. Turn demo mode (password list on the sign-in page) off when the demo period ends.

6. If something goes wrong

Problem	Cause / fix
No Review / → button for an approver	Wrong province in the header, or the previous step was not done, or this person already decided another tier
"Please confirm your identity"	Sign-in older than 10 minutes; re-enter the password
Voucher shows On hold	Payments → **Holds** tab shows why (stay order, mutation, lease hold)
Explorer sees no progress strip	The Land Officer has not created the acquisition yet
Request does not become Completed	The acquisition is not at Handover yet, or the hand-over kind was not Acquisition (use **Mark land ready**)

Problem	Cause / fix
Stay order held nothing	Parcels / landowners were not added on the legal case
Map is blank	No internet for tiles: Basemap → None
Codes have different numbers	Records were created before; carry on
New screens missing	Build and restart not done after `git pull` / migrate